SEMINAR REPORT

A seminar Report on

AI Privacy Gateway

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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ABSTRACT

Using LLMs in enterprise applications can lead to exposure of confidential Personally Identifiable Information (PII) to the external cloud environment, resulting in privacy risks and non-compliance to regulations such as GDPR and HIPAA. Static redaction technique eliminates all detected PII without consideration of semantic context leading to reduced AI response utility.

To address the privacy-utility dilemma, this work proposes AI Privacy Gateway, a local edge-based approach placed between the client applications and the cloud LLMs. The approach involves identification of sensitive entities and evaluation of their relevance for a specific task performed with the help of a local Named Entity Recognition (NER) model and Small Language Model (SLM). Task-relevant PII entities are retained, whereas other entities are replaced by meaningful, reversible tokens in a local dictionary. The sanitized prompt is sent to the cloud LLM, and a post-processing module automatically maps the original entities back to the generated answer. Experimental analysis demonstrates that the proposed approach reaches 93%+ entity detection accuracy, 8%+ adversarial data recovery and maintains 80%+ answer utility compared to full-masking baselines.

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CHAPTER 1
INTRODUCTION

Artificial Intelligence (AI) and Large Language Models (LLMs) are used in various fields of human activity, from education to healthcare, from business to customer service, and from information retrieval to answering queries. However, the problem is that users often provide personal information, such as age, name, address, phone number, email, financial information, and many others, when using such systems. The usage of such information in prompts may lead to possible privacy and data security issues if prompts are shared with other entities, such as AI models.

The AI Privacy Gateway is a system intended to prevent the dissemination of private information and keep personal data secure. The tool allows you to separate irrelevant and valuable information for the AI model. While the AI Privacy Gateway operates as an intermediate service, the input prompt is analyzed, and personally identifiable information (PII) is masked.

The system is designed to operate on a context-based privacy paradigm, which means that all personal information that can be removed without damaging the request's context will be removed. For instance, when a user sends an inquiry about staying awake at work, the system will mask the PII data, such as the place of work, with little or no impact on the request's context. Such a method allows the AI Privacy Gateway to operate efficiently without reducing the quality of responses provided by an external AI service.

The AI Privacy Gateway can be equipped with various tools, such as Named Entity Recognition, local LLMs, contextual analysis, and reversible masking. The system can use these tools to identify sensitive data and keep track of the necessary information to restore it when an external AI service provides an answer.

The ultimate goal of the AI Privacy Gateway is to ensure privacy and data confidentiality while maintaining the quality of the results provided by external AI systems. Therefore, it is essential to evaluate and measure the efficiency of the AI Privacy Gateway through metrics such as accuracy in detecting PII, privacy protection, response quality, and speed. In this way, the AI Privacy Gateway will ensure the safety of personal information and reduce the exposure risk when using an external AI service.

CHAPTER 2
LITERATURE REVIEW

1. PII-Bench: Evaluating Query-Aware Privacy Protection Systems

This work introduces PII-Bench, a benchmark for assessing efficacy of query-aware privacy protection systems. The experiments demonstrate the difficulty of distinguishing between personally identifiable information (PII) and irrelevant information with respect to a given query.

2. Portcullis: A Scalable and Verifiable Privacy Gateway for Third-Party LLM Inference

This work introduces Portcullis, a privacy gateway for third-party large language model inference. It aims to sanitize or replace sensitive information in prompts before they are sent to LLMs, and to securely deliver appropriate information back to the user.

3. Casper: Prompt Sanitization for Protecting User Privacy in Web-Based Large Language Models

This work introduces Casper, a system for protecting user privacy in web-based large language models. It analyses prompts for potentially sensitive information, and applies rules and name entity recognition to sanitize them before they are sent to web-based AI assistants.

4. A Secure API Gateway Framework for Enterprise Applications

This work presents a secure API gateway framework for enterprise applications. The framework includes OAuth 2.0 authentication, JWT-based authorization, rate limiting, and AI-powered anomaly detection to help protect enterprise applications and improve the security of their APIs.

5. Federated Learning with Secure API Gateways for Enhancing Privacy in Distributed AI Systems

This work presents an approach to enhance the privacy of distributed artificial intelligence systems through the use of secure API gateways and differential privacy. It investigates the application of these techniques in the context of federated learning, and the impact on information leakage from the model updates.6. AI-Powered Integrated with Encoding Mechanism Enhancing Privacy, Security, and Performance for IoT Ecosystem

6. Operationalizing data minimization for privacy-preserving prompting of large language models

This article explores the ways to minimize personal information in prompts to AI while ensuring that the response still addresses the user's request. The authors suggest using such actions as retaining, abstracting, and redacting to find the most privacy-preserving prompt version that would still help achieve the goal.

7. Legal Guardian: A Privacy-preserving Framework for Secure Integration of Large Language Models into Legal Practice

This framework is designed to obscure personal information in prompts to a large language model while integrating AI into legal practice. It is focused on detecting personal information and masking it with consistent placeholders before the prompt is sent to an external AI. The redacted version is then replaced with the response, unless the initial prompt is reconstructed by the legal AI. However, LegalGuardian does not consider whether the personal information is relevant to the case before removing it.

8. CAPID: Context-Aware PII Detection for Question-Answering Systems

is a research paper that proposes using locally fine-tuned language models for detecting personally identifiable information, classifying its type, and determining relevance to the user's question. The technology aims to keep the task-relevant information intact while masking the irrelevant personal data to achieve a better balance between privacy and answer quality.

9. Local-First Privacy Firewall for AI Applications: A Risk-Adaptive Framework for Protecting Sensitive User Prompts

is a research paper proposing a browser-based privacy firewall solution, which can detect sensitive information locally and score its risk level. The tool aims to allow prompts to be masked, replaced, or flagged depending on the detected value and risk to the user before being sent to an external service for processing.

10. Representation-Level Privacy for Cloud LLM Inference: A Client-Side Perturbation Framework for Reducing Raw Prompt Exposure

is a research paper proposing an alternative approach to reducing raw prompt exposure by transforming request representations on the client before sending them to a compatible cloud-based language model. The technology aims to help achieve better privacy by analysing the prompt's impact, but it also considers the quality and extra computational costs of such processing.

11. Privacy-Preserving Audit Gateway for LLM Applications: Detecting Prompt Threats While Minimizing Stored User Content

is a research paper that introduces a privacy-preserving audit gateway concept as a way to detect prompt injection threats and the exposure of sensitive data. The tool aims to reduce the amount of prompt content stored in readable form while allowing auditing and investigation into suspicious activity by analysing the risk level and applying the necessary transformations.

12. Extracting Training Data from Large Language Models

This research demonstrates that large language models can reveal memorized training data, including potentially sensitive information. It highlights the privacy risks associated with deploying large language models and the need for effective mechanisms to prevent unintended disclosure.

Table No 2.1: Literature review**:**

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Benchmarking query-aware PII detection and relevance judgments.	Models and prompt strategies are compared; specific names are not listed.	PII-Bench: 2,842 samples; 7 PII types; 55 categories.	Test whether systems detect PII and decide what to keep or mask for a query.	Relevance judgments lag behind PII detection; performance drops on long, dense, multi-person prompts. More medical and financial cases are needed.
2	Privacy gateway using anonymization and verifiable response restoration.	Anonymization, substitution, and response restoration; specific tools/models are not stated.	Not stated in the supplied notes.	Mask sensitive details before third-party LLM processing and restore appropriate values in the response.	Uniform masking ignores query relevance; context-aware decisions remain a research opportunity.
3	Prompt sanitization using rules, NER, and local processing.	Rule-based checks; Named Entity Recognition (NER); local processing. Specific library/model not stated.	Web-based LLM prompts; no named benchmark is stated.	Detect and sanitize sensitive prompt details before sending them to an online AI service.	May hide details needed for the task because the approach does not explicitly assess relevance.
4	Risk-adaptive browser prompt filtering with user review.	TypeScript; Web Extensions API; ONNX Runtime Web; regular expressions; token classifier; browser storage.	Proposed metrics: precision, recall, F1, leakage rate, and task utility.	Detect sensitive data locally and allow, warn, redact, or replace it before external submission.	Depends on detector and policy quality; limited browser coverage; does not cover all inference attacks or compromised devices.
5	Client-side prompt representation transformation and perturbation.	Python; transformer encoder/tokenizer; PyTorch; custom inference server; embedding perturbation.	No fixed dataset; proposed tests cover leakage attacks, task quality, bandwidth, and latency.	Perturb a local prompt representation before sending it to a compatible cloud inference server.	Representations may leak information, while stronger perturbation can harm quality. Requires server adaptation; may not support commercial APIs.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
6	Prompt-threat detection with minimal audit logs and fingerprint matching.	Python; FastAPI; text classifier; cryptographic hashes; vector representations.	No fixed dataset; proposed metrics include precision, recall, F1, attack recall, latency, log size, and audit usefulness.	Detect prompt injection and sensitive-data exposure while storing minimal event metadata instead of readable prompts.	Fingerprints can leak data or cause false matches. Detection does not replace tool authorization or output validation.
7	Enterprise API gateway with access control and ML anomaly detection.	OAuth 2.0; JSON Web Tokens (JWT); ML-based anomaly detection.	Examples: banking, healthcare, and IoT APIs; no specific benchmark stated.	Secure microservice access and detect unauthorized activity or DDoS attempts.	Adds latency and requires training data; false positives/negatives may occur. Does not inspect PII and can be a single point of failure.
8	Federated learning with a secure gateway and differential privacy.	Kong Gateway; OAuth 2.0; JWT; TLS; differential privacy; S-FLCF model; zero-trust concepts.	CIFAR-10 and MIMIC-III; 100 simulated edge nodes, according to the notes.	Protect federated learning while raw data stays local and model updates use a secure gateway.	Privacy adds accuracy/communication costs and latency. Standard benchmarking and scalability testing are needed; some attack figures are illustrative.
9	Reversible Replacement Encoding for IoT data and intrusion detection.	Replacement Encoding; Random Forest; DNN; Genetic Algorithm; secret-key-based encoding.	CICIoT2023; reported accuracy: Random Forest 94.81%, DNN 90.16% with feature selection.	Camouflage sensitive IoT data while supporting model training, intrusion detection, and preprocessing.	May be vulnerable to cryptanalysis; collisions and key-management risks remain. Tests focus on IoT traffic, not text prompts.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Freeze-Then-Search prompt optimization using RETAIN, ABSTRACT, and REDACT.	Fine-tuned Qwen2.5-7B; GPT-5 and other LLMs; priority-queue search; RETAIN/ABSTRACT/REDACT actions.	ShareGPT, WildChat, CaseHOLD, and MedQA; nine LLMs in the supplied notes.	Find the least revealing prompt that still produces a useful answer.	Requires many LLM checks and relies on model-based utility judgments; models may overuse abstraction and reveal too much.
11	Mask-and-unmask pipeline using entity detection and consistent placeholders.	GLiNER; Qwen2.5-14B; placeholder dictionary; local LLM processing.	50 synthetic immigration-law prompts; about 460 entities, according to the notes.	Protect legal-client PII with placeholders and restore original values in responses when needed.	Tested only on synthetic data in one legal domain. Masks all PII regardless of relevance; proposed hybrid detector is not implemented.
12	Fine-tuned small language models for PII detection, typing, and relevance labels.	Llama-3.1-8B LoRA; Llama-3.2-3B LoRA; GPT-4.1-mini; synthetic-data generation and relevance labeling.	2,307 synthetic QA samples across 15 PII types; 150 real Reddit excerpts.	Detect PII, classify it, and decide whether it should be kept or masked using a local model.	Weaker on long prompts and limited-context questions; relevance is binary; domain rules and broader real-world coverage are limited.

CHAPTER 3
INFORMATION/CHAPTERS

1. Overview of AI Privacy Gateway:

An AI Privacy Gateway is a piece of software placed between the users and the external AI systems. It detects personal information in users' prompts and obscures non-essential details, ensuring the privacy of personal data and helping to maintain the quality of responses provided by the AI service.

2. Personally Identifiable Information:

Overall, personally identifiable information (PII) encompasses any information that can pertain to a particular person. For instance, PII includes email addresses, phone numbers, physical addresses, financial data, and sensitive medical records. An AI Privacy Gateway scans through the provided text for any PII data and determines what information should be removed or replaced in order to prevent the disclosure of personal details.

3. Need for AI Privacy Gateway:

While scanning through the text for PII data, an AI Privacy Gateway can eliminate some details that may be useful for generating a high-quality response. That is why there is a need for context-aware privacy protection that would identify relevant information and leave it intact while masking the PII data, thus, reducing the amount of personal data that would be processed by an AI system.

4. Working Principle:

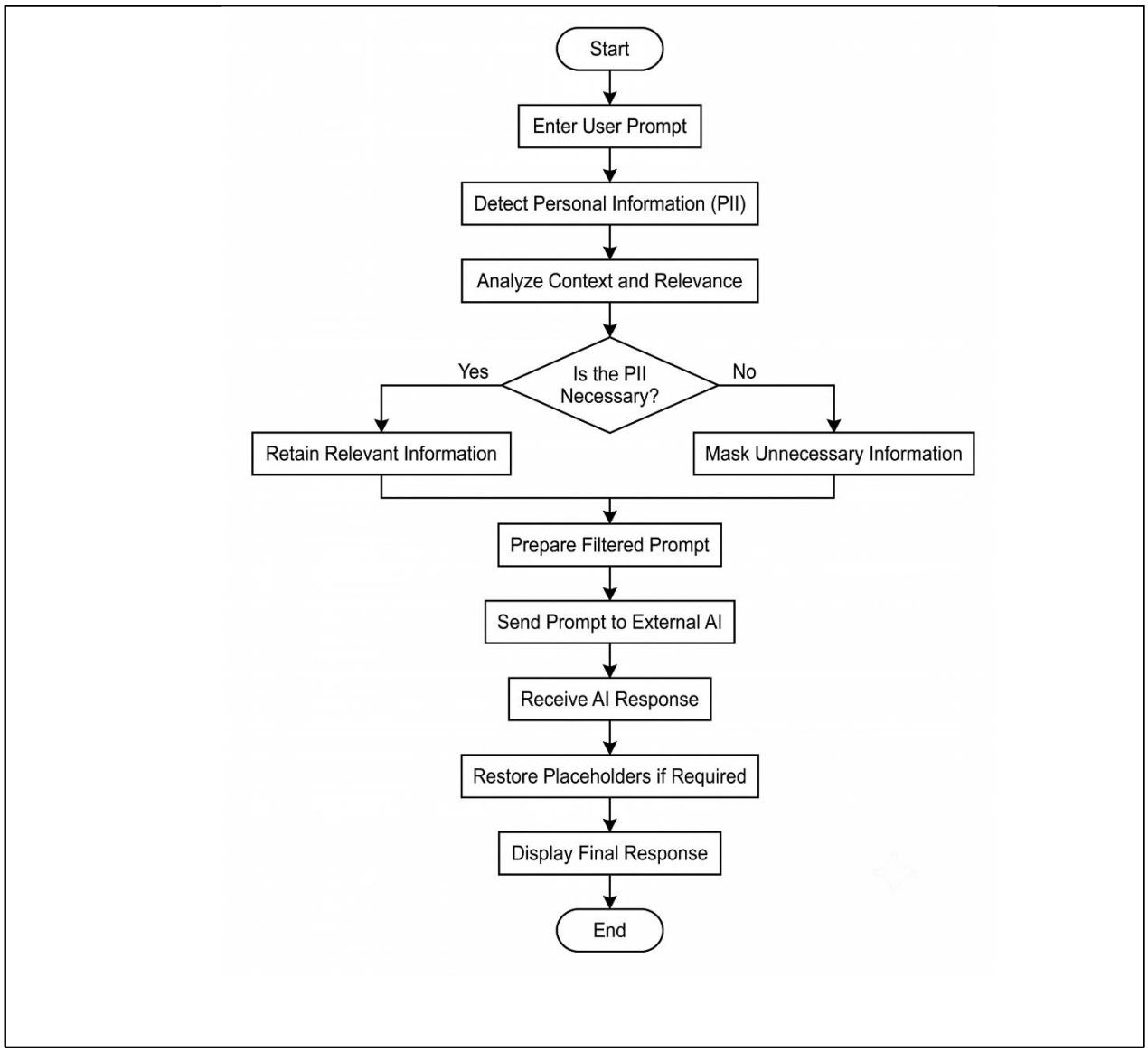


Fig 3.1: Working of AI Privacy Gateway

The working of the AI Privacy Gateway consists of the following steps:

1. Start: In the starting, the process of protecting user data is initiated when the user wants to interact with an external AI system.

2. Enter User Prompt: At this step, the user enters a question or a prompt, containing the information which an AI system needs to generate a response.

3. Detect Personal Information (PII): At this step, the gateway scans the prompt to identify the personal information like name, phone number, email address, and location.

4. Analyze Context and Relevance: Finally, the gateway analyses the detected information to determine if any information needs to be redacted or not.

5. Decide whether PII is necessary: If the information is relevant it is retained, if it is irrelevant it is masked to protect the user's privacy.

6. Retain or Mask Information: Relevant information is retained and unnecessary information is masked.

7. Prepare filtered Prompt: The gateway uses the relevant information and masked information to prepare a privacy protected prompt.

8. Send Prompt to External AI: The prompt is sent to the external AI for processing.

9. Receive AI Response: The external AI processes the prompt and returns an AI response to the gateway.

10. Restore Placeholders if required: If required the gateway restores the placeholders using a protected mapping.

11. Display Final Response: The final response is displayed to the user with reduced exposure of personal information.

12. End: End of process after the final response is delivered to the user.

5. System Architecture and Main Components:

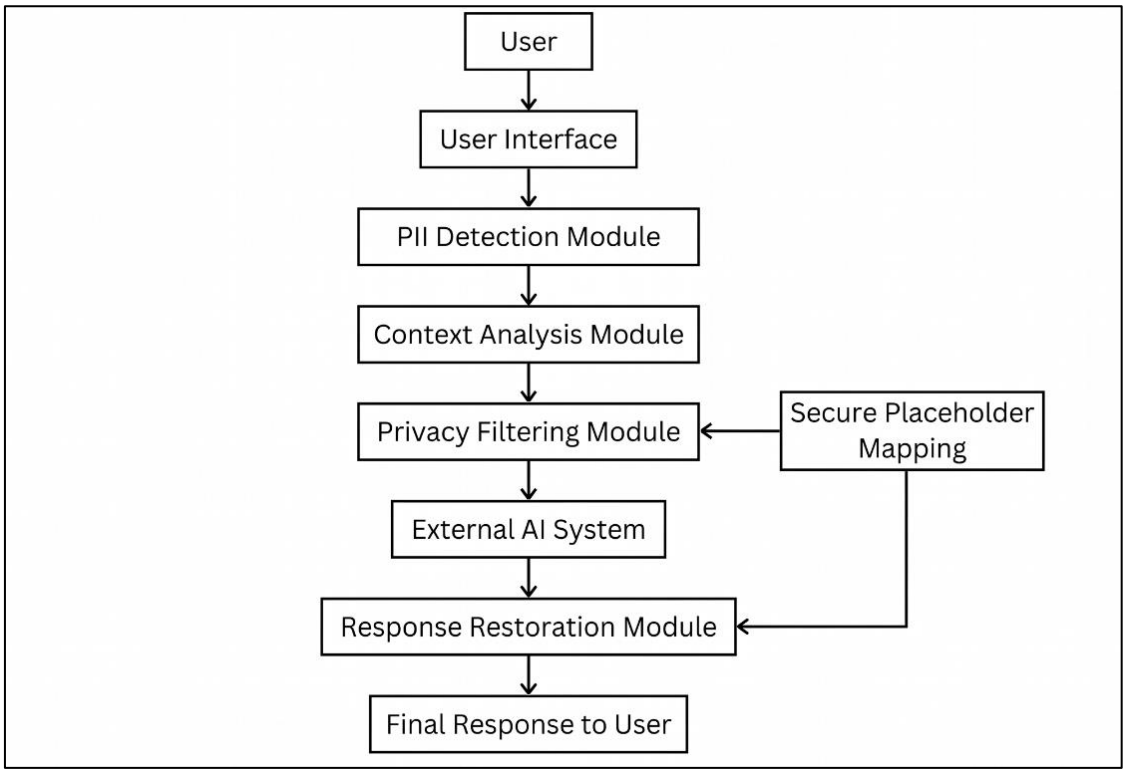


Fig 3.2: Architecture of AI Privacy Gateway

The components are: User Interface, PII Detection Module, Context Analysis Module, Privacy Filtering Module, External AI Interface, and Response Restoration Module. All of them perform their function to detect any PII, determine what data to hide and how, communicate with the outside, and present the final response to the user.

6. Technologies and Techniques Used:

Python can be used to build the gateway. Regular expressions will detect phone numbers and email addresses, while Named Entity Recognition will identify the names and places. Language models can be utilized to perform context analysis, and placeholder mapping may replace the sensitive details with labels like [PERSON_1] for the final technology selection based on the project implementation and existing resources.

7. Example of the PII Filtering:

"My name is Rahul. I am a warehouse supervisor with a back pain. I live in Pune. How can I reduce fatigue at work?"

In this example, the gateway can remove the name and the place of residence as they are not relevant to the question. The result will be the following: "I am a warehouse supervisor with a back pain. How can I reduce fatigue at work?" Thus, the prompt will still contain the necessary personal identifiers to get an adequate response.

9. Privacy – Preserving Processing of the Response:

After the external large language model processes the request, the gateway can replace any of the identifiers with the placeholders. When the response is sent back to the user, the placeholders should be replaced with the corresponding data, if there are any. It should be noted that the privacy-preserving processing of the request will only minimize the risks of personal data exposure; it will not eliminate them. Therefore, the responses containing the identifiers should be masked to prevent the unauthorized access to the personally identifiable information.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Enhanced privacy due to the masking of unnecessary information prior to the request being sent to an external AI system.
2. Context preservation through the filtering mechanism, while retaining the necessary information to the question.
3. Ensured the response quality is not impacted due to the relevance of the masked information provided to the AI application.
4. Reduced exposure of information to the external AI application.
5. The solution can be easily integrated into different AI applications that may use different privacy-preserving mechanisms and/or APIs.

4.2 Disadvantages:

1. The gateway may fail to identify some personal information.
2. The gateway may misidentify ordinary text as sensitive information.
3. Incorrect masking of important details might degrade the quality of the response provided by the AI assistant.
4. The process may take more time to analyze and mask PII data compared to systems without this feature.
5. In cases where the context is not understood correctly by the system, the filtered question or response provided by the AI system might not be relevant.
6. Proper measures should be employed to protect the mapping between the placeholder tags and the actual values to avoid exposing the original private information in case of any security breach.

CHAPTER 5
CONCLUSION

The AI Privacy Gateway offers a solution to the privacy concerns associated with using external AI systems. This tool can identify personally identifiable information (PII), evaluate the pertinence of data to the request, and mask irrelevant information in the prompt to the third-party AI service.

The AI Privacy Gateway can be a great addition to any AI service. By addressing the issues of PII, context assessment, privacy masking, and response reinstallation, the tool can be utilized to ensure privacy without compromising on the content of the response. While the implementation of such a gateway poses several challenges, addressing these issues can lead to a privacy-centric experience with AI services, which would increase the adoption rate of these technologies. In summary, an AI Privacy Gateway can be a great addition to any AI service by offering an additional layer of protection and ensuring that the user's data is handled responsibly.

CHAPTER 6
PROJECT INITIATION

6.1 Description of Problem Statement:

The development of Artificial Intelligence (AI) applications presents new threats to the privacy of personal information. Users provide prompts to external AI services, which may contain PII such as their names, contact details, emails, location, and health information. At the same time, the use of AI services requires users to provide prompts containing personal information that may subsequently be exploited. Conventional data masking techniques may eliminate all private data from prompts, thus reducing the usefulness of AI results. Therefore, a privacy gateway is needed that can identify PII, determine which personal data to mask, and allow generating prompts that still yield valuable results. The AI Privacy Gateway would be placed in between the user and an external AI service, scanning input prompts for PII, analyzing the relevance of each identified item, and masking the information. The gateway would ensure that only the necessary data are passed to the AI and reconstruct the response by replacing any masked items in the response with the corresponding placeholders.

Contextual analysis can make the AI Privacy Gateway effective by recognizing irrelevant PII and ensuring that only the necessary data are masked. This way, the privacy gateway would balance the need for personal data protection and the need for users to receive valuable AI results. For instance, when a user asks for directions to a specific address, the AI Privacy Gateway would mask the address, name, and contact details of the user while leaving the request intact. In another scenario, the AI Privacy Gateway would recognize the user's location data if they ask for local weather forecasts.

6.2 Scope of the Project:

The purpose of this project is to design an AI Privacy Gateway that will safeguard the personal information of a user interacting with external AI applications. The system will detect Personally Identifiable Information (PII) in a user's prompts and analyze the relevance of every piece of information to a specific query. It will make sure to keep relevant information and remove or mask the other details before sending the prompts to an outside system. The AI gateway can also put back the values in a response if they are needed and appropriate. The

project aims to ensure that the user's data exposure is unnecessary while still getting the required answers from the external applications.

Objectives of the Project:

1. To find out all the personally identifiable information from the user prompt;

2. To examine the connection between the found data and the user's problem;

3. To mask or delete all the unnecessary personal data from the prompt;

4. To keep the required PII to ensure the results' quality and usefulness;

5. To create a reliable placeholder system that would allow restoring the necessary information when needed;

6. To minimize the risk of providing users with excessive data.

6.3 Stakeholders:

The stakeholders of this AI Privacy Gateway are those who will use the product, develop and maintain the product, or be affected by its development and use.

1. End Users: These are people who will use AI applications and seek to protect their private information when using AI.

2. Developers: This group includes people responsible for developing, implementing, and maintaining the Gateway.

3. AI Service Providers: This category includes organizations providing third-party AI models or APIs for processing filtered prompts.

4. System Administrators: These are individuals who manage the gateway, ensure security, and perform maintenance tasks.

5. Organizations: This category includes universities, companies, and organizations that use AI applications and need more privacy and security when storing data.

6. Project Guide and Evaluators: These are faculty members who guide the project and evaluate its performance.

6.4 Platform / Equipment / Resources:

1. Platform:

The AI privacy gateway can be built on Windows or Linux operating systems through a Python programming language. Any code editor can be applied for developing and testing the software, for example, Visual Studio Code. For practical implementation, the system can be connected to the external artificial intelligence via an API to execute filtered prompts.

2. Equipment:

- **Computer or Laptop:** For developing and running the application.
- **Processor:** A standard modern processor suitable for Python development.
- **RAM:** Minimum 8 GB recommended; additional memory may be required for running local language models.
- **Internet Connection:** Required for accessing external AI services and downloading libraries.
- **Storage:** Sufficient space for source code, required libraries, and model files, if used.

3. Resources:

Resources	Name	Purpose
Operating System	Windows 11	Host and run development tools, editors, and local testing environment.
Programming Language	Python 3.11	Primary language used to implement application code, scripts, and tests.
Code Editor	Visual Studio Code	Code editing, debugging, and extension support for development.
PII Detection Tool	Microsoft Presidio	Detect and classify personally identifiable information in text.
Pattern Matching	Python re Module	Perform regular-expression based pattern matching and extraction.
NLP Library	spaCy	Natural language processing tasks such as tokenization, parsing, and entity recognition.
Language Model	Qwen2.5-7B-Instruct	Instruction-following large language model for generation or assistance tasks.
AI Service	OpenAI API	Access hosted models and APIs for additional AI capabilities and integrations.

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960174	388494	Ansari Mohammad Sami Zahid Husain
2	24610960176	388496	Bagul Aditya Yuvraj
3	24610960178	388498	Bhadane Madhuri Ajay
4	24610960228	388536	Palve Pavitra Kiran

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388494	388496	388498	388536
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : **Shraddha S Hajari**

(Course Expert/s)